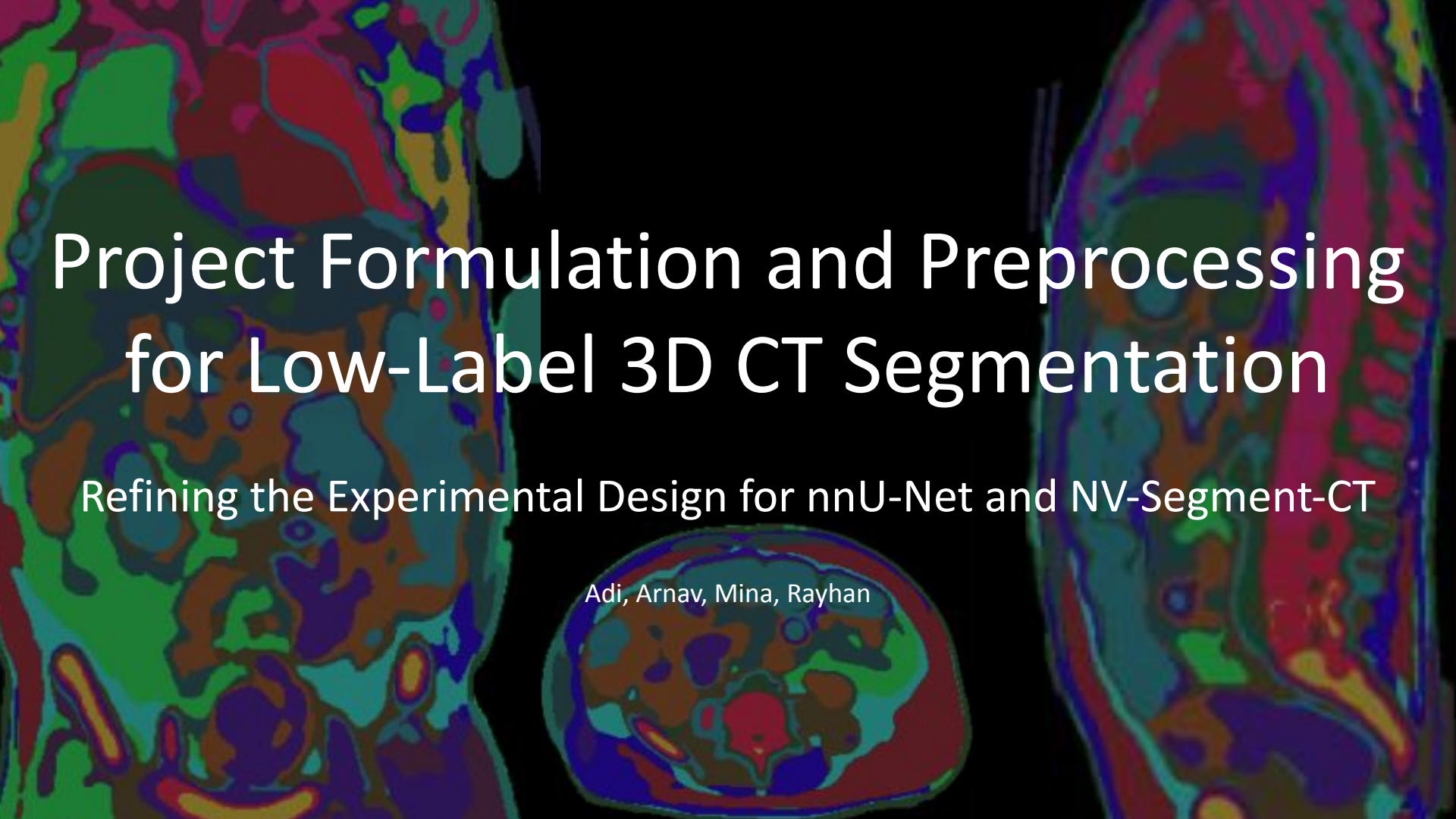




Project Formulation and Preprocessing for Low-Label 3D CT Segmentation

Refining the Experimental Design for nnU-Net and NV-Segment-CT

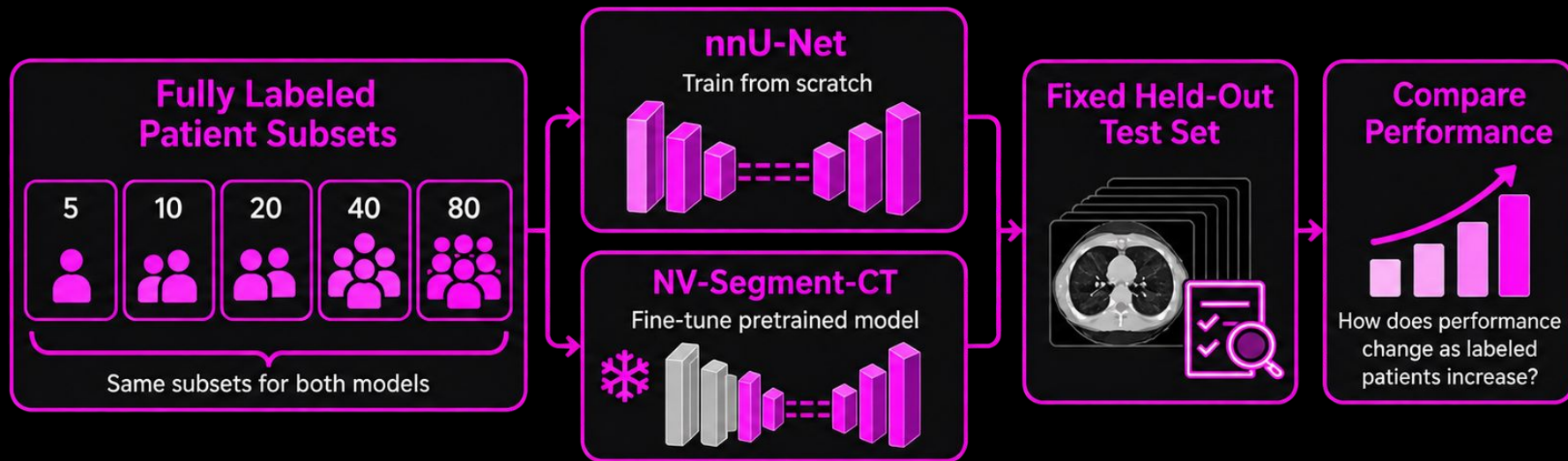
Adi, Arnav, Mina, Rayhan

Project Recap

- 3D CT segmentation requires costly expert annotation
- Goal: determine when pretraining reduces labeling requirements

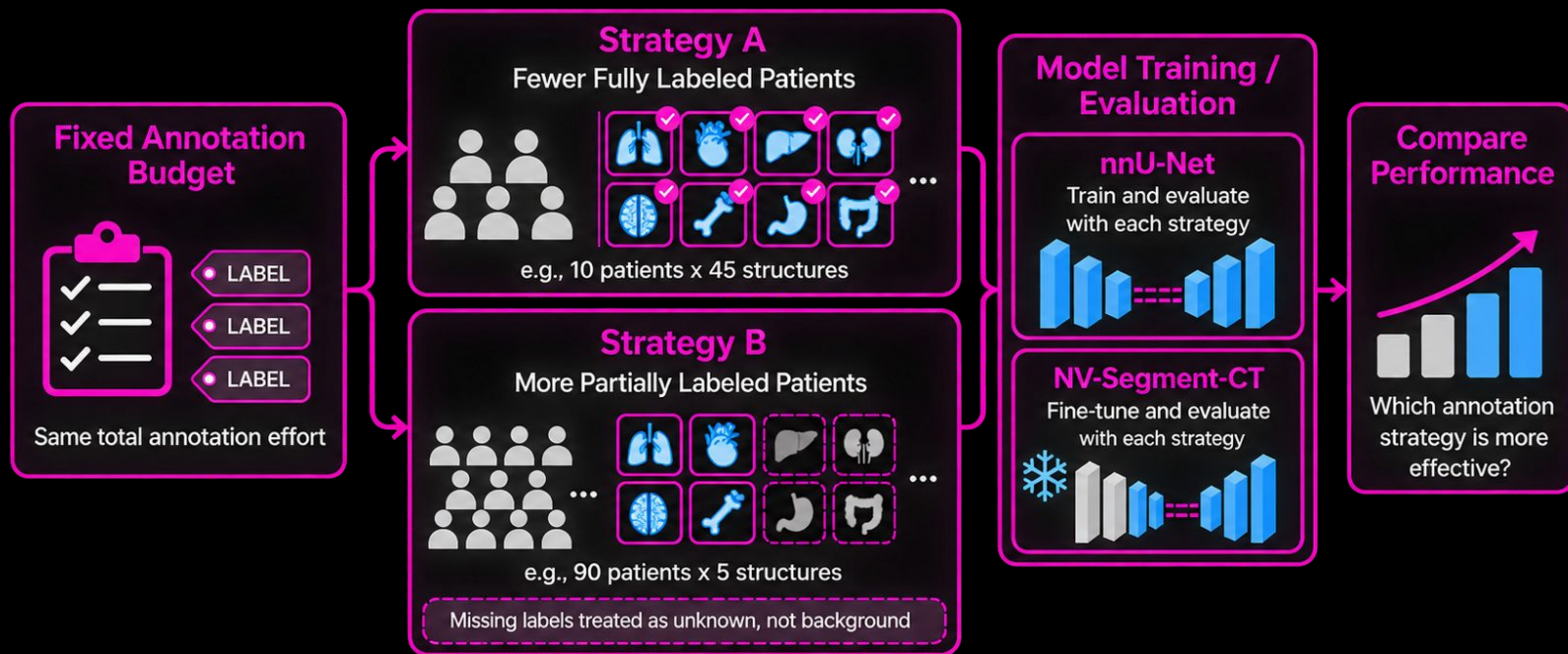


Specific Aim 1 ~ Patient-Level Label Scarcity



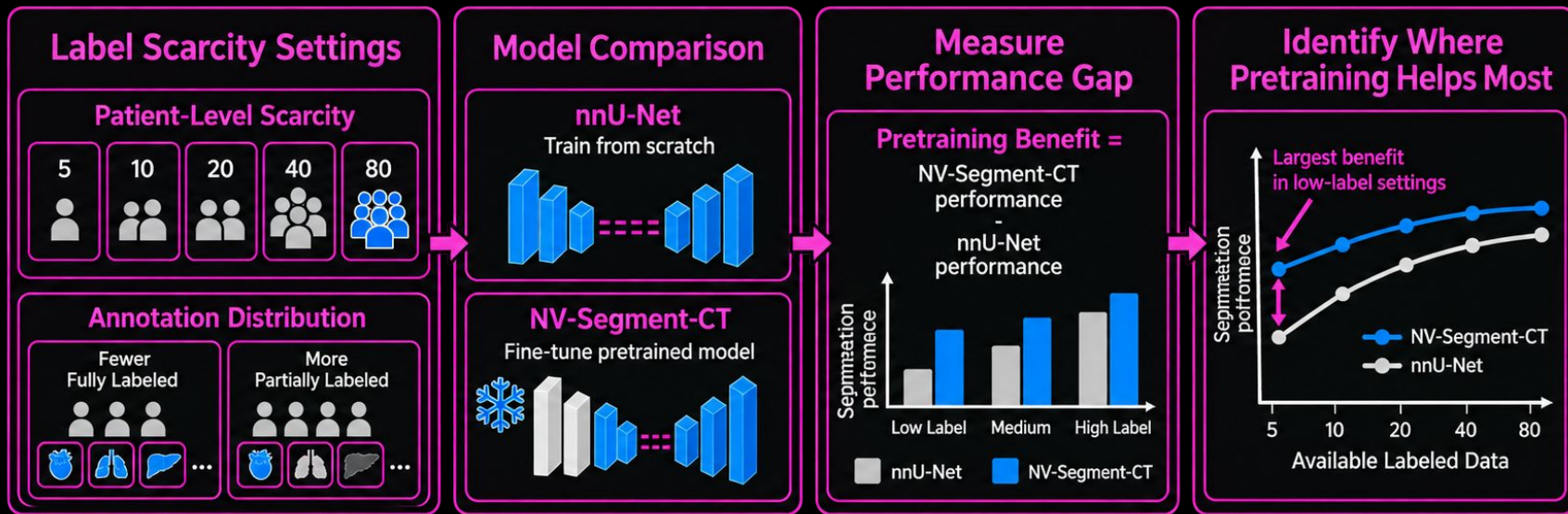
Takeaway: Compare both models across 5–80 fully labeled patients using the same held-out test set.

Specific Aim 2 ~ Annotation Distribution



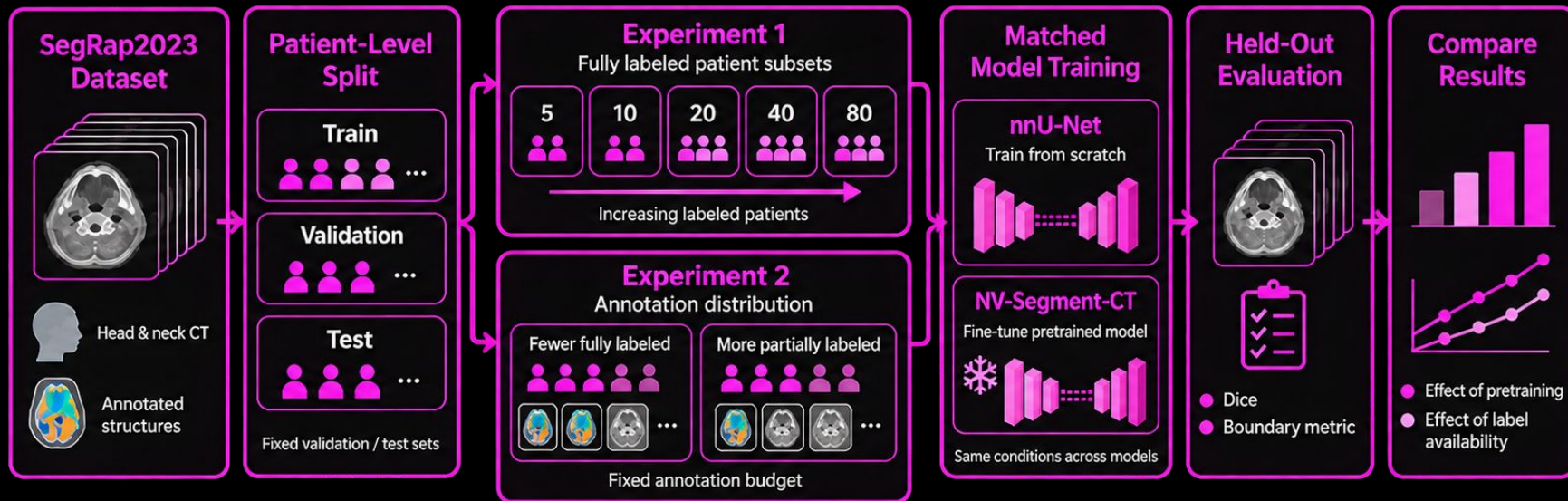
Takeaway: Under a fixed annotation budget, compare fewer fully labeled patients with more partially labeled patients.

Specific Aim 3 ~ Value of Pretraining Under Label Scarcity



Takeaway: Measure where the performance advantage of pretraining is largest across scarcity conditions.

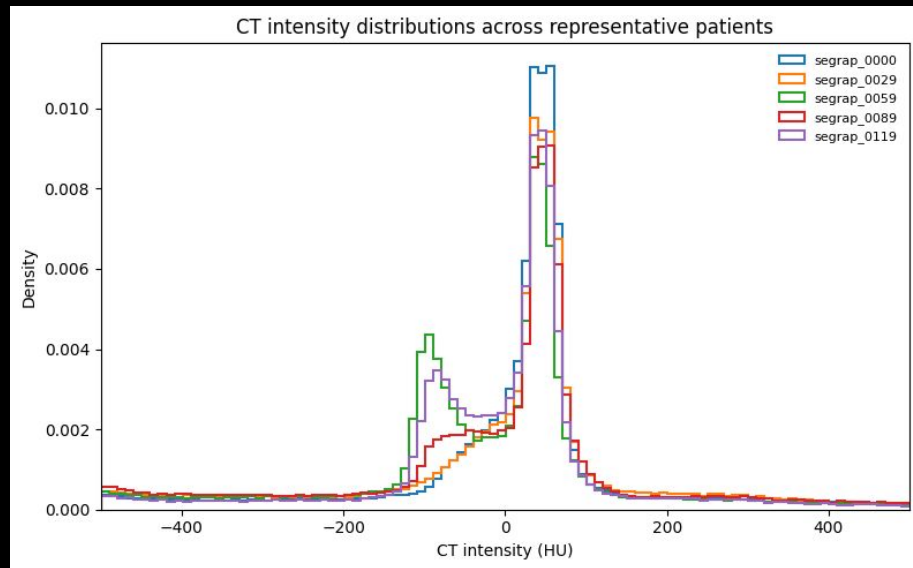
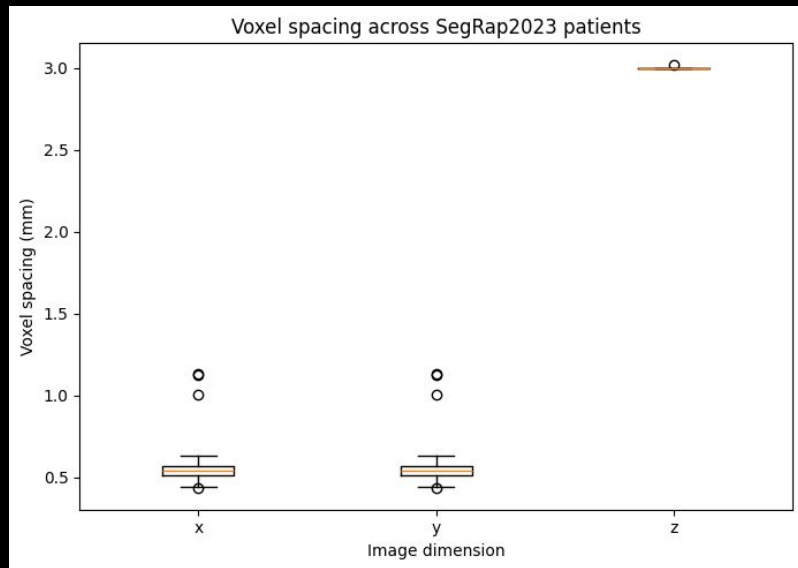
Experimental Design



Takeaway: Both models are evaluated under matched data splits, target structures, and test conditions.

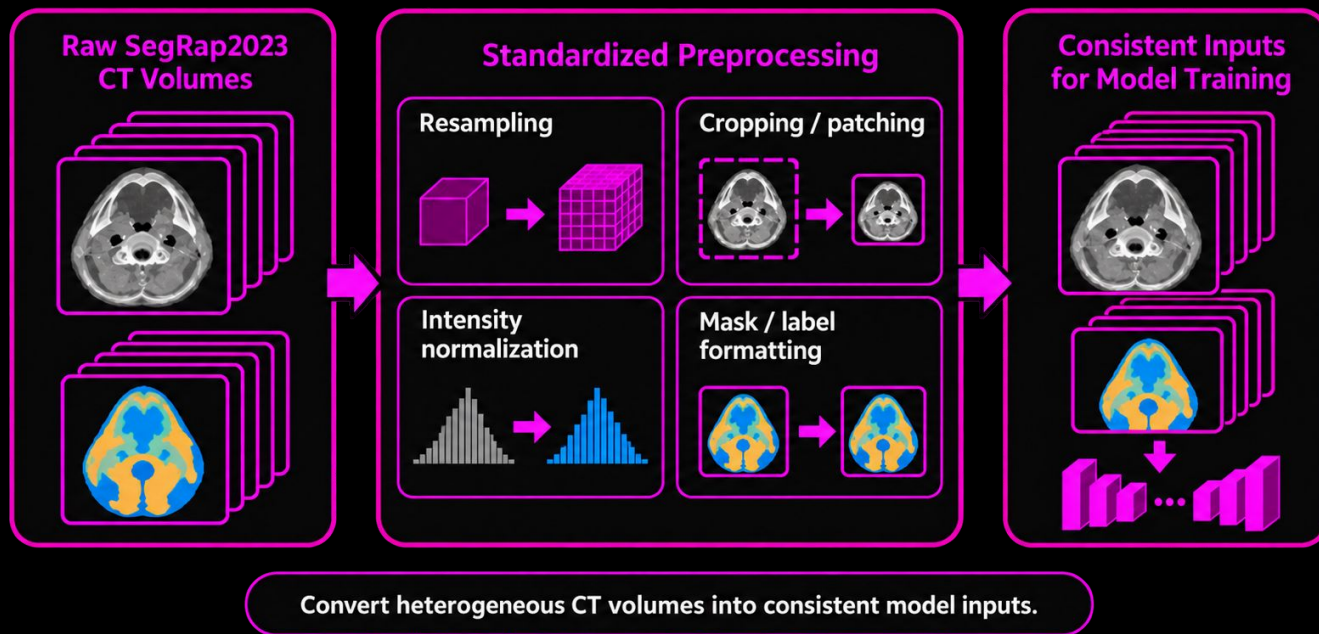
Evaluation Metrics: Dice Score + Boundary-Sensitive Metric

What Our EDA Revealed



Key Takeaway: Variation in scan geometry and intensity distributions motivates standardized preprocessing before model training.

Preprocessing: What Needs to Be Standardized?



Takeaway: Spatial, intensity, and label inconsistencies must be standardized before training.

Why We Chose nnU-Net as the Baseline

Manual / Custom 3D U-Net Pipeline

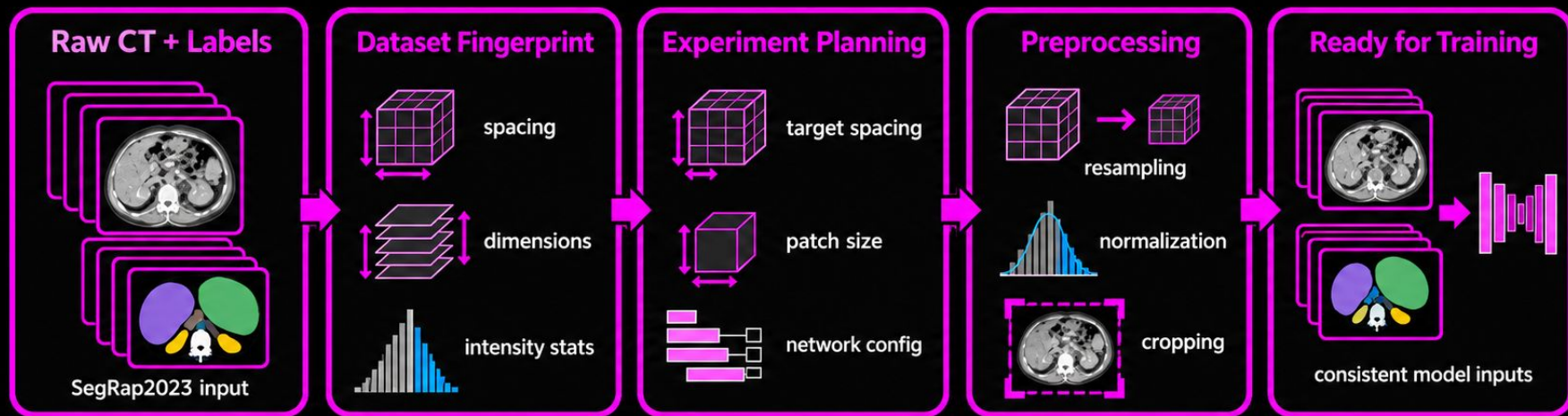
- Manual preprocessing decisions
- Manual spacing/patch size
- More arbitrary tuning

nnU-Net

- Automatic data-driven preprocessing
- Dataset-adaptive configuration
- Standardized medical segmentation baseline

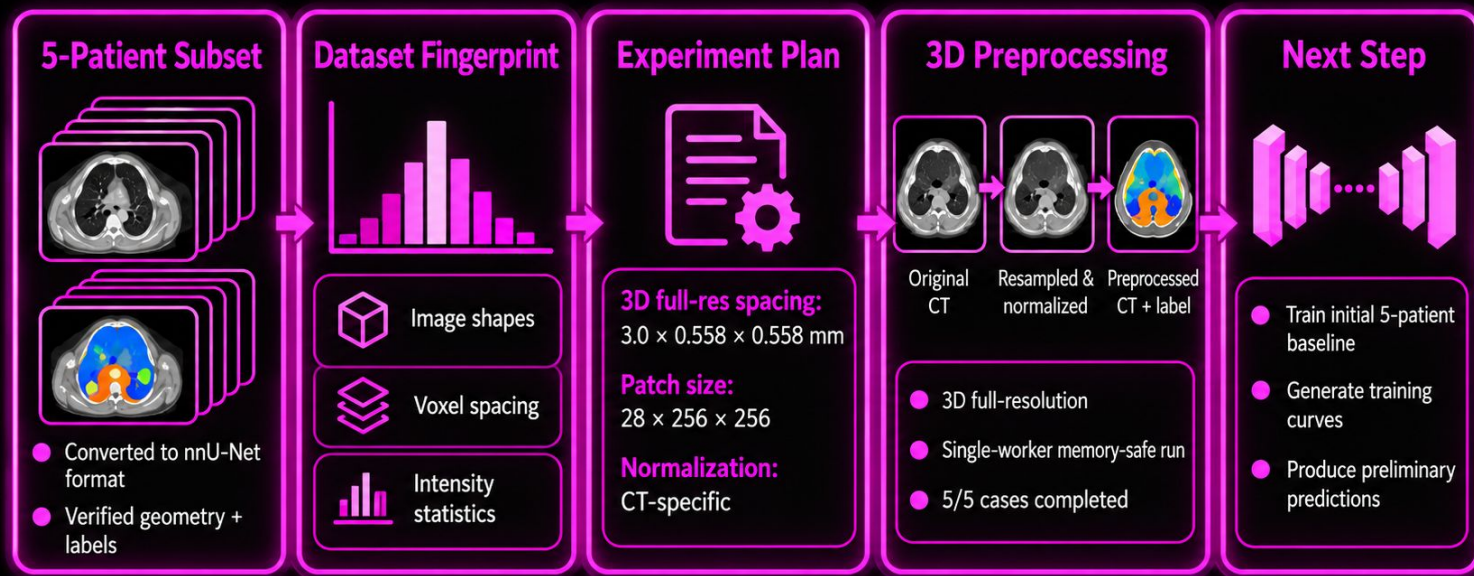
Why it matters: we want performance differences to reflect the models, not arbitrary preprocessing choices

nnU-Net Preprocessing Pipeline



Takeaway: nnU-Net converts dataset characteristics into a standardized preprocessing and training configuration.

Initial nnU-Net Preprocessing Results



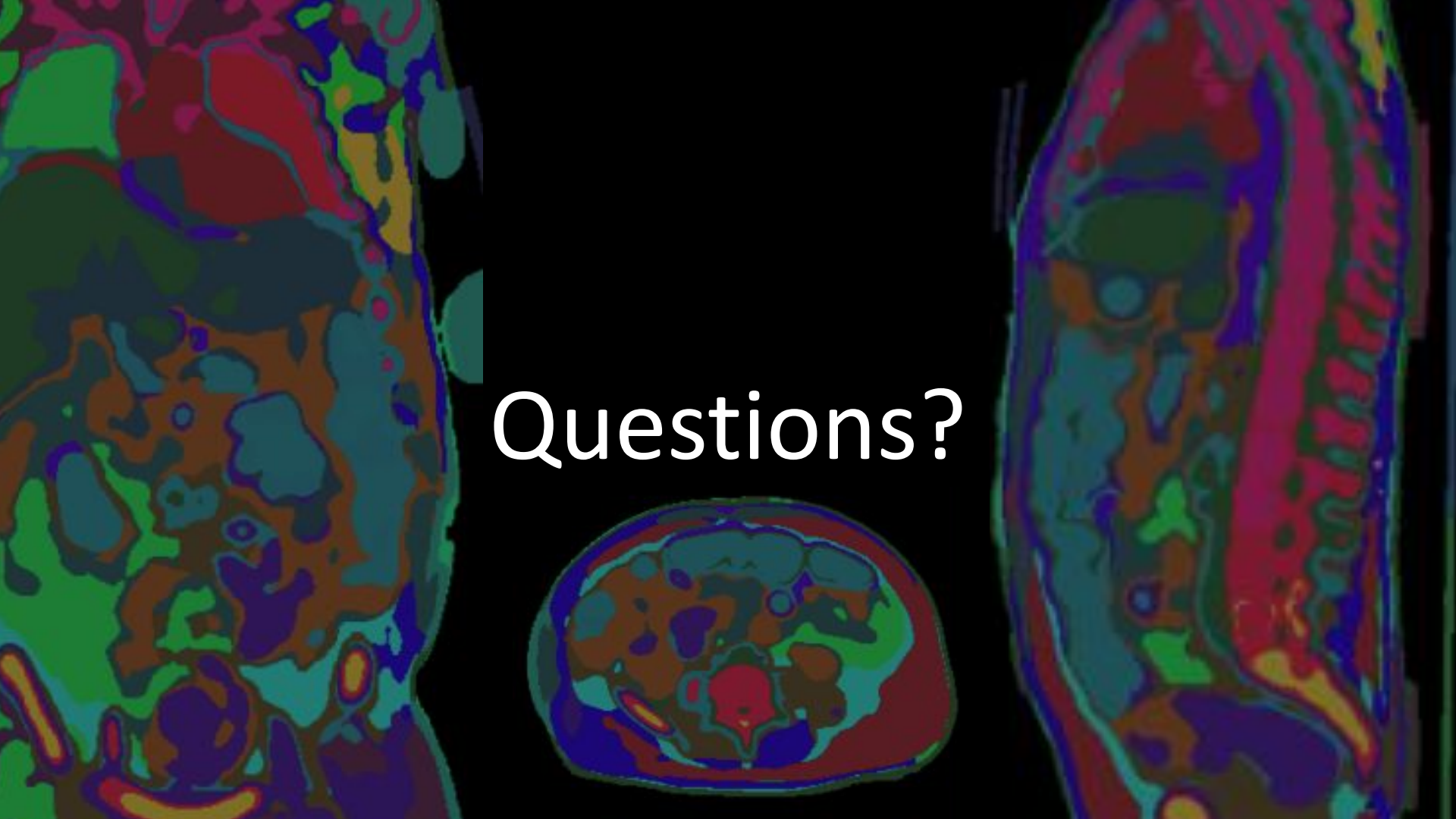
Takeaway: The nnU-Net preprocessing pipeline was successfully validated on the initial 5-patient subset and is ready for baseline training.

Progress Summary

- Completed EDA and characterized key dataset variability
- Refined project questions, specific aims, and experimental design
- Selected nnU-Net as the baseline and converted a 5-patient subset
- Completed fingerprinting, experiment planning, and 3D preprocessing for 5/5 patients

Plan for Next Week

- Train the initial 5-patient nnU-Net baseline
- Generate preliminary training curves and segmentation predictions
- Verify the full training → inference → evaluation pipeline
- Finalize evaluation metrics for the first baseline run

The image features three MRI scans of a human torso, likely a female, showing internal organs. The scans are color-coded: red for soft tissue, green for fat, blue for fluid, and yellow for bone. The top-left scan is a coronal view, the top-right is a sagittal view, and the bottom-center is an axial view. The word "Questions?" is written in white text across the center of the image.

Questions?